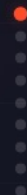


Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and
Hardware Attestation Wins

arxiv.org/abs/2504.04715



01

The Model Substitution Problem

Providers face strong economic incentives to covertly substitute cheaper alternatives — quantized versions, smaller models, or different architectures — while maintaining pricing. The black-box nature of APIs means users have no guarantee.

01

Smaller Model

Serve 8B parameters instead of 70B to cut GPU costs

02

Quantized Variant

FP8 / INT8 precision instead of FP16 — near-invisible quality drop

03

Fine-tuned Version

Silently updated weights change model behavior over time

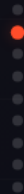
04

Different Family

An entirely different architecture behind the same API endpoint

$H_0: P_{\text{actual}}(y|x) = P_{\text{spec}}(y|x)$ vs. $H_1: P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$

Formalized as hypothesis testing — can an auditor distinguish the specified model's distribution from the actual one?



FP8 Quantization Preserves Most Accuracy

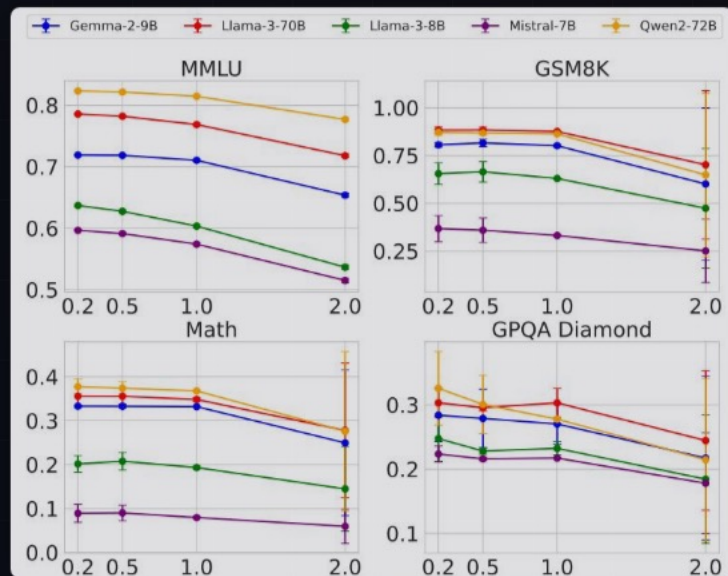
Performance drops are small — often within error bars — making substitution nearly undetectable via benchmarks alone.

MODEL	MMLU	GSM8K	MATH	GPQA DIAMOND
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19



Higher Temperature Amplifies Gaps but Increases Noise

Temperature scaling experiments (0.2, 0.5, 1.0, 2.0) show that higher temperatures degrade all models, further obscuring substitution signals.



Benchmark accuracy vs. temperature for five model families across four tasks – from the original study

Log-Prob Fingerprinting Fails Due to Inference Nondeterminism

The same Llama-3-8B model produces different log probabilities across software versions (transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2) and GPU hardware (A100 vs H100).

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Log-probability variation across frameworks and hardware – indistinguishable from actual substitution



Randomized Routing Makes Detection Query-Prohibitive

Providers can actively evade audits through adversarial strategies that exploit the statistical limits of black-box testing.



Randomized Substitution

Routing only fraction p of queries to the cheaper model. As $p \rightarrow 0$, the mixed distribution converges to the specified model, requiring exponentially more queries.

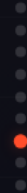
Benchmark Evasion

Detecting known benchmark prompts and routing them to the genuine model while serving substitutes for regular traffic.

Software-Only Auditing Methods Are Fundamentally Unreliable

Three distinct failure modes render every software-based detection approach ineffective in practice.

METHOD	CORE WEAKNESS	PRACTICAL LIMITATION
Text-Based Statistical Tests (MMD)	Requires impractically large query budgets to achieve significance	Fails against subtle or quantized substitutions where outputs are nearly identical
Log-Probability Fingerprinting	Production inference nondeterminism across software versions and GPU hardware	Cannot distinguish legitimate infrastructure variation from actual model swaps
Adversarial Countermeasures	Randomized routing + benchmark evasion actively defeat detection	Provider can reduce substitution rate p to make detection query-prohibitive



TEEs Provide Provable Cryptographic Model Integrity

Trusted Execution Environments — hardware-secured enclaves — offer cryptographic attestation that the correct model weights are loaded and inference code is running.

Provable

Cryptographic attestation guarantees model identity — no statistical sampling required

Efficient

Only modest performance overhead vs. full inference — unlike ZKPs which are computationally prohibitive for large models

Robust

Immune to adversarial evasion — hardware-level guarantees cannot be circumvented by routing tricks or benchmark detection

Deployable Today

NVIDIA confidential computing on H100 GPUs provides a practical deployment path for TEE-based inference verification

Software Methods

Statistical tests, log-prob fingerprinting, and adversarial countermeasures all fail — fundamentally limited by the black-box API boundary

Key Takeaways

Close the verification gap with hardware attestation.

- 1 **Model substitution is economically motivated and practically undetectable** via software — providers can cut GPU costs with near-zero impact on observable quality.
- 2 **FP8 quantization barely affects benchmark scores** — most drops fall within error bars across MMLU, GSM8K, MATH, and GPQA Diamond.
- 3 **Inference nondeterminism kills log-probability fingerprinting** — framework versions and GPU hardware produce variation indistinguishable from substitution.
- 4 **Randomized substitution defeats statistical tests** — as the substitution rate p decreases, detection requires exponentially more queries.
- 5 **TEEs are the only currently viable path to verifiable LLM APIs** — hardware attestation provides provable, efficient, and robust integrity guarantees.

Adopt TEE-based attestation for LLM inference — the tooling exists today on NVIDIA H100 confidential computing.